SEVA FIRST HACKATHON REGISTRATION

Team Registration:

Team Name: **VelaiVaasal**

Participant 1 Name: **SHARAN ANANDAN SUJATHA**

Date of Birth: 17-01-2009

Gender: MALE

Class/Year: BE. CSE / 1st YEAR

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Date of Birth: 19.02.2009

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Participant 3 Name: **VASUDEV R**

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Gender: MALE

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(MAXIMUM PARTICIPANTS:3 MEMBERS,ONE GIRL STUDENT)

ID Card(Upload): **Scanned Copy to be Uploaded Here**







Options	Your answer
Title of the solution	VelaiVaasal: Your Gateway from Classroom to Career – A Multilingual Learning-to-Placement Companion

Theme	**Shikshit & Kaushal Bharat**
Problem Statement	Many students, especially first-generation and rural learners, finish their courses without a clear learning path, interview practice or any contact with employers, so their skills and available jobs never meet.
Track	**Product / Technology Solution**
What problem your are saying? Maximum 500 words allowed.	India produces millions of graduates every year, yet many are not ready for their first job. The India Skills Report 2026 (ETS with CII, AICTE, AIU and Taggd), based on more than one lakh candidates and over 1,000 employers, puts national employability at 56.35%. This means roughly 44 in every 100 assessed graduates were not job-ready. The picture is weaker in vocational streams: only 45.95% of ITI candidates and 32.92% of polytechnic diploma holders were assessed as employable. Students get stuck at three points. First, a learning gap: study material is scattered and is not matched to the skills employers actually ask for, so learners do not know what to study next. Second, a readiness gap: students get few chances to practise aptitude tests, build a recruiter-ready resume or face a mock interview with honest feedback. Third, an access gap: rural and first-generation learners rarely meet mentors or recruiters, and most tools are in English and need a fast connection. The result is that talent exists everywhere but access does not. A capable student in a small town may never learn how to present their skills, while employers struggle to find candidates who are prepared. Colleges with small or overstretched placement cells cannot give every student personal guidance. Our problem statement is therefore how to give every learner, in their own language and on an ordinary phone, a clear path from classroom to first job.
What the proposed Solution Maximum 1000 words allowed.	VelaiVaasal (Tamil for “gateway to work”) is a mobile-friendly web platform that measures where a student stands, shows the gap to a chosen job role, and gives a personalised plan and interview practice to close it, in the student's own language (Tamil, Hindi and English at launch). Its core is an Evidence-to-Readiness Engine: assessment evidence, competency profile, role requirements, skill-gap diagnosis, then an action plan with reassessment. 1. AI Readiness Analyzer: A baseline assessment of technical skills, aptitude, communication and

	problem solving gives evidence-backed scores instead of self-reported claims. 2. Skill Gap Detector: The student's competencies are compared with the requirements of the target role. The platform shows strengths, missing skills and priorities. 3. Mock Interview Coach: Role-specific technical and HR practice, by voice or text, with feedback on relevance, clarity and correctness. 4. Personalised Roadmap: A weekly plan built from the measured gaps. It adapts as the student improves and reassesses. 5. Company Readiness: Demonstrated skills are compared with selected job-role profiles to give a role-readiness score. It is a preparation guide, not a hiring guarantee. 6. TPO Intelligence: Placement officers see department trends, common gaps and students who need support, so training time goes where it is needed. Design for access: pages are light for low-end phones and weak networks, voice input is offered for learners who are more comfortable speaking than typing, and colleges can onboard whole batches at once. Later extensions: Resume Studio, and Opportunity Match to internships, jobs and volunteer mentors from partner employers and NGOs, are planned after the pilot. Outcome: students know what to learn and can show measurable progress, colleges get data to plan training, and placement officers get actionable insights.
Who is affected? and where does the problem occur Maximum 300 words allowed.	Primary users are final-year and pre-final-year students of arts and science colleges, engineering colleges, polytechnics and ITIs. The students who benefit most are first-generation learners, learners from rural and small-town colleges, and women who have fewer networks and mentors. The problem occurs in colleges and training institutes across India, particularly where placement cells are small or where students study in regional languages. It also affects college placement officers, who cannot give personal guidance to hundreds of students, and employers and NGOs that want to hire or mentor local talent but cannot easily find prepared candidates. The India Skills Report 2026 shows that employability is rising in Tier-2 and Tier-3 cities, which means the talent is there and what is missing is structured preparation and access.

Who is innovative or different about your solution? Maximum 300 words allowed.	Most placement tools are built for English-speaking students in big cities and treat learning, practice and placement tracking as separate products. VelaiVaasal is different in five ways. Evidence, not self-reporting: skill gaps come from an assessment-based competency profile compared with the requirements of the chosen role, so the same student can have a different readiness score for Data Analyst and for Java Developer. One connected loop: assessment, diagnosis, personalised practice and reassessment work together, so a mock interview feeds the weekly plan instead of standing alone. Prioritised by measured weaknesses: the roadmap starts with the gaps that matter most for the target role, not a generic course list. Language first, built for access: the interface, questions and feedback work in Tamil, Hindi and English, with voice input and light pages for low-end phones and weak networks. Insights for placement officers: department competency trends and support flags, not only placement results.
How your solution does occurs? Maximum 300 words allowed.	Step 1, Onboard: A student signs up, picks a target job role and a preferred language. Step 2, Assess: A baseline assessment covers technical skills, aptitude, communication and problem solving, and produces a competency profile. Step 3, Diagnose: The profile is compared with the target role's requirements. Gaps are ranked by how much they matter for the role, and a role-readiness score is shown. Step 4, Practise: In the mock interview module the student answers by voice or text. Speech recognition converts speech to text, and the answer is scored for relevance, structure and confidence with suggestions to improve. Scores are stored so progress can be tracked. Step 5, Improve and reassess: A weekly roadmap of lessons and mini-tasks targets the biggest gaps. The student reassesses and the plan adapts. Placement officers see trends on a dashboard. Technology (planned): React web app (PWA), Python (FastAPI) back end, an LLM API with Tamil and Hindi speech-to-text, and PostgreSQL (Supabase). The hackathon prototype uses React, Flask and SQLite, a rule-based scorer in place of the LLM, and browser speech recognition.
What stage your solution is at?	**Prototype**

Idea Prototype Working Prototype Pilot Deployed	
Additional notes on your stage Maximum 300 words allowed.	VelaiVaasal is at prototype stage. The hackathon build is a working web app with sample data. A student chooses one of three roles (Junior Data Analyst, Customer Support Executive or Accounts Assistant), completes a four-level skill check, and receives a gap-ranked roadmap with lessons, mini-tasks and a job-readiness score. The mock interview coach asks nine HR and role questions, accepts voice or typed answers, scores relevance, structure and confidence, and saves progress. The interface and questions work in English, Tamil and Hindi. Seven automated API tests pass and the full flow was tested in a browser. Limits: the skill check is self-rated and is not yet a measured assessment. Interview scoring is a rule-based stand-in for the planned LLM. The prototype runs on Flask and SQLite instead of FastAPI and PostgreSQL. Student login, the TPO dashboard, Company Readiness, Resume Studio and Opportunity Match are not built yet. It has not been tested with real students. Next steps: add baseline assessments and a competency profile, student login and LLM-based feedback, then run a pilot with partner colleges to collect feedback on the roadmap and the mock interview.
Who will benefit or what impact will create? Maximum 300 words allowed.	Direct beneficiaries are students, who gain a clear learning path, practice and access to opportunities. First-generation, rural and women learners gain the most because the platform replaces the personal mentoring they often lack. Colleges and placement cells benefit through a dashboard and a ready-made preparation tool. Employers and NGOs benefit from better-prepared candidates and a simple way to reach local talent. Year-one targets for the pilot (our estimates): 5,000 students trained across 10 partner colleges at about 500 learners each; 10,000 mock interviews completed at two per student; and 500 internships or jobs facilitated, which assumes 10% of learners reach an outcome. Wider impact: higher confidence, shorter job searches, and more local employment, which supports the Shikshit & Kaushal Bharat theme. For students: clearer direction and visible progress. For colleges: data-driven training. For TPOs: actionable insights. We will judge success by improvement in assessment scores, fewer unaddressed priority gaps, learning-plan completion and the number of students placement

	officers can support.
What is the estimated cost of your solution **Maximum 300 words allowed.**	Indicative estimate for the year-one pilot (5,000 learners): about Rs. 14 lakh in total, or around Rs. 280 per learner. These are our planning estimates and will be refined during the pilot. Cloud hosting and database: about Rs. 1.2 lakh. AI and speech services for about 10,000 mock interviews: about Rs. 3 lakh. Content curation and Tamil/Hindi localisation: about Rs. 2.5 lakh. Pilot operations, mentor coordination and college onboarding: about Rs. 2 lakh. Part-time development and maintenance: about Rs. 4 lakh. Contingency of about 10%: about Rs. 1.3 lakh. Costs per learner fall as more colleges join, because the platform and content are reused.
How can be the solution implemented and scaled? **Maximum 300 words allowed.**	Implementation happens in three phases. Phase 1, Hackathon: a working prototype with role selection, a skill check, a skill-gap roadmap and a mock interview coach in English, Tamil and Hindi. Phase 2, Pilot: add student login, baseline assessments and the competency profile, then run with two to three partner colleges for one semester and collect feedback on the roadmap and the usefulness of the mock interview. Phase 3, Scale: add the TPO dashboard, more languages, NGO partners and recruiter connections, and extend to polytechnics and ITIs, where employability gaps are widest. Scaling is practical because the platform is cloud-based and content is reusable. A new college can be onboarded with a batch upload of students, and the per-learner cost stays low. A train-the-trainer model with placement officers and student volunteers allows growth without a large central team.
What evidence you have?	**Other (published national data and prototype)**
Describe your evidence? **Maximum 300 words allowed**	Our evidence so far comes from published national research and our prototype. The India Skills Report 2026 (ETS with CII, AICTE, AIU and Taggd; about 1.03 lakh candidates across 20 states and 3 UTs, and 1,000+ employers) reports overall employability of 56.35%, with ITI candidates at 45.95% and polytechnic diploma holders at 32.92%. The same report notes that Tier-2 and Tier-3 cities are emerging as employability hubs, which shows where preparation support can have the most

	effect. Our hackathon prototype demonstrates the skill-gap roadmap and mock interview flow with sample data, and its automated tests pass. We have not yet run our own student survey or pilot. Phase 2 will collect user feedback, usage data and placement outcomes to test our assumptions.
Who could adopt or implement the solution? Maximum 300 words allowed	Colleges and universities: placement cells and training and placement officers can adopt it for whole batches. Polytechnics and ITIs: vocational institutes where employability gaps are largest. State and central skilling programmes: skilling missions and similar initiatives can use VelaiVaasal as a delivery tool. NGOs and CSR teams: organisations that run training or mentoring can use it to reach and support learners. Employers and recruiters: companies hiring freshers can use it to find prepared local candidates.
What support do you need to take the solution forward? Maximum 300 words allowed	Pilot partners: two to three colleges willing to onboard students and give feedback. Mentors and employers: volunteers, NGOs and companies who can offer mock interview sessions, internships and entry-level roles. Funding and credits: pilot funding of about Rs. 14 lakh, or equivalent cloud and AI service credits. Content and language experts: educators and translators to review Tamil and Hindi lessons and interview feedback. Guidance: mentorship from the hackathon panel on data privacy, student safety and responsible use of AI.
Concept Document	 Seva_Fest_Hackathon_ Placement_Education.
Other files or documents	**https://github.com/Sharan-4223/VelaiVasal**
Video Link (optional)	